

EXP:4	EDA-Data Cleaning
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CODE:

```
import pandas as pd
```

```
data = {  
    'Gender': ['Male', 'Female', 'Male', 'Female', 'Male',  
              'Female', 'Male', 'Female', 'Male', 'Female'],  
    'Married': ['Yes', 'No', 'Yes', 'Yes', 'No',  
              'Yes', 'No', 'Yes', 'Yes', 'No'],  
    'Dependents': [1, 0, 2, 1, '3+', 0, 2, 1, 0, 3],  
    'Education': ['Graduate', 'Graduate', 'Not Graduate', 'Graduate',  
                 'Graduate', 'Not Graduate', 'Graduate', 'Graduate',  
                 'Not Graduate', 'Graduate'],  
    'Self_Employed': ['No', 'No', 'Yes', 'No', 'Yes',  
                    'No', 'No', 'Yes', 'No', 'No'],  
    'ApplicantIncome': [5000, 3000, 7000, 4500, 8000,  
                       3500, 6000, 4000, 5500, 3200],  
    'CoapplicantIncome': [1500, 0, 2000, 1000, 2500,  
                         500, 1200, 0, 1800, 700],  
    'LoanAmount': [150, 100, 200, 120, 250,  
                  110, 180, 130, 160, 105],  
    'Loan_Amount_Term': [360, 360, 360, 240, 360,  
                       360, 180, 360, 240, 360],  
    'Credit_History': [1, 1, 0, 1, 1,
```

```
        1, 0, 1, 1, 1],
        'Property_Area': ['Urban', 'Rural', 'Semiurban', 'Urban', 'Rural',
                           'Semiurban', 'Urban', 'Rural', 'Semiurban', 'Urban']
    }

df = pd.DataFrame(data)

print("First 5 Rows:")
display(df.head())

print("\nLast 5 Rows:")
display(df.tail())

print("\nDataset Information:")
df.info()

print("\nDescriptive Statistics:")
display(df.describe())

print("\nColumn Names:")
print(df.columns.tolist())

categorical_cols = [
    'Gender', 'Married', 'Dependents',
    'Education', 'Self_Employed', 'Property_Area'
]
```

```
print("\nUnique Values:")

for col in categorical_cols:
    print(f"\n{col}:")
    print(df[col].unique())

df['Dependents'] = df['Dependents'].replace('3+', 3)
df['Dependents'] = pd.to_numeric(df['Dependents'], errors='coerce')

print("\nValue Counts of Dependents:")
print(df['Dependents'].value_counts())

cols = ['ApplicantIncome', 'CoapplicantIncome', 'LoanAmount']

for col in cols:
    print(f"\nStatistics for {col}:")
    print(f"Mean: {df[col].mean():.2f}")
    print(f"Median: {df[col].median():.2f}")
    print(f"Mode: {df[col].mode().values}")
    print(f"Range: {df[col].max() - df[col].min():.2f}")
    print(f"Variance: {df[col].var():.2f}")
    print(f"Standard Deviation: {df[col].std():.2f}")
```

OUTPUT:

First 5 Rows:

	Gender	Married	Dependents	Education	Self_Employed	ApplicantIncome	CoapplicantIncome	LoanAmount
0	Male	Yes	1	Graduate	No	5000	1500	
1	Female	No	0	Graduate	No	3000	0	
2	Male	Yes	2	Not Graduate	Yes	7000	2000	
3	Female	Yes	1	Graduate	No	4500	1000	
4	Male	No	3+	Graduate	Yes	8000	2500	

Last 5 Rows:

	Gender	Married	Dependents	Education	Self_Employed	ApplicantIncome	CoapplicantIncome	LoanAmount
5	Female	Yes	0	Not Graduate	No	3500	500	
6	Male	No	2	Graduate	No	6000	1200	
7	Female	Yes	1	Graduate	Yes	4000	0	
8	Male	Yes	0	Not Graduate	No	5500	1800	
9	Female	No	3	Graduate	No	3200	700	

Dataset Information:

```
<class 'pandas.core.frame.DataFrame'>
```

RangeIndex: 10 entries, 0 to 9

Data columns (total 11 columns):

#	Column	Non-Null Count	Dtype
0	Gender	10 non-null	object
1	Married	10 non-null	object
2	Dependents	10 non-null	object
3	Education	10 non-null	object
4	Self_Employed	10 non-null	object
5	ApplicantIncome	10 non-null	int64
6	CoapplicantIncome	10 non-null	int64
7	LoanAmount	10 non-null	int64
8	Loan_Amount_Term	10 non-null	int64
9	Credit_History	10 non-null	int64
10	Property_Area	10 non-null	object

dtypes: int64(5), object(6)

memory usage: 1012.0+ bytes

Descriptive Statistics:

	ApplicantIncome	CoapplicantIncome	LoanAmount	Loan_Amount_Term	Credit_History
count	10.000000	10.000000	10.000000	10.000000	10.000000
mean	4970.000000	1120.000000	150.500000	318.000000	0.800000
std	1666.033213	841.691419	48.215373	69.570109	0.421637
min	3000.000000	0.000000	100.000000	180.000000	0.000000
25%	3625.000000	550.000000	112.500000	270.000000	1.000000
50%	4750.000000	1100.000000	140.000000	360.000000	1.000000
75%	5875.000000	1725.000000	175.000000	360.000000	1.000000
max	8000.000000	2500.000000	250.000000	360.000000	1.000000

Column Names:

['Gender', 'Married', 'Dependents', 'Education', 'Self_Employed', 'ApplicantIncome', 'CoapplicantI

Unique Values:

Gender:

['Male' 'Female']

Married:

['Yes' 'No']

Dependents:

[1 0 2 '3+' 3]

Education:

['Graduate' 'Not Graduate']

Self_Employed:

['No' 'Yes']

Property_Area:

['Urban' 'Rural' 'Semiurban']

Value Counts of Dependents:

Dependents

1 3

0 3

2 2

3 2

Name: count, dtype: int64

Statistics for ApplicantIncome:

Mean: 4970.00

Statistics for ApplicantIncome:

Mean: 4970.00

Median: 4750.00

Mode: [3000 3200 3500 4000 4500 5000 5500 6000 7000 8000]

Range: 5000.00

Variance: 2775666.67

Standard Deviation: 1666.03

Statistics for CoapplicantIncome:

Mean: 1120.00

Median: 1100.00

Mode: [0]

Range: 2500.00

Variance: 708444.44

Standard Deviation: 841.69

Statistics for LoanAmount:

Mean: 150.50

Median: 140.00

Mode: [100 105 110 120 130 150 160 180 200 250]

Range: 150.00

Variance: 2324.72

Standard Deviation: 48.22

```
/tmp/ipykernel_11634/2807943612.py:56: FutureWarning: Downcasting behavior in `replace` is deprecated
df['Dependents'] = df['Dependents'].replace('3+', 3)
```